Risheek Bajaj

AI & Software Systems Engineer | Data Scientist | Analytics & Automation Architect

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AI & Software Systems Engineer and Data Scientist with ~3 years of experience architecting scalable software applications, automated enterprise workflows, end-to-end ETL data pipelines, and production machine learning models across Healthcare Tech, Manufacturing, and Digital Marketing. Proven track record of cutting inventory carrying costs by 18%, boosting campaign ROI by 18%, and eliminating 500+ hours of annual manual operations. Backed by a strong portfolio of multiple deployed full-stack AI/ML and software engineering projects—including an LLM-powered HR Resume Intelligence web app (Streamlit), a 0.92 ROC-AUC Financial Fraud Detection Engine, a PyTorch/CNN Visual Search System, and an automated SQL Server Data Warehouse. Distinguished by elite global certifications and credentials including US Fellowships in Applied AI & Data Science (WorldQuant University), MeitY (Govt. of India), NPTEL (IIT) Cloud & Database Systems, HackerRank SQL (Intermediate), two peer-reviewed Springer ML publications, and an MCA with Distinction.

TECHNICAL SKILLS

- **Software Engineering & Backend:** Python (Pandas, NumPy, Decorators, Generators, Scikit-Learn, PyTorch), SQL, NoSQL (MongoDB), Google Apps Script, Flask, Streamlit, Plotly Dash, REST APIs, AppSheet, Git/GitHub.
- **Business Intelligence & Analytics:** Power BI, DAX, Streamlit, Plotly Dash, Tableau, Automated KPI Frameworks, Exploratory Data Analysis (EDA), Statistical Modeling, Advanced Excel.
- **AI, Machine Learning & Deep Learning:** Large Language Models (LLMs), Computer Vision (CNNs, ResNet, YOLO, TensorFlow Lite), PyTorch, Scikit-Learn, Predictive Modeling, Time-Series Forecasting (Prophet, ARIMA/ARMA, GARCH), GANs, SMOTE, A/B Testing.
- **Engineering & Automation:** Data Engineering, ETL Pipelines, AppSheet No-Code Development, Database Administration, DPDP Act 2023 Security & Compliance, Hardware Deployment

EXPERIENCE

Automation Specialist & Data Systems Engineer – Project Based **May 2026 – Present**
Development Consortium | Bahraich / New Delhi, India
Project on Breastfeeding and New born Care (PBNC) — Public Health Tech & Data Infrastructure

- **Enterprise Workflow Automation:** Architected and deployed full-scale patient and program management applications using Google Apps Script and AppSheet, automating multi-stage data ingestion pipelines and outbound telephony workflows.
- **Database Engineering & High Availability:** Administered and secured the central healthcare database; engineered automated validation safeguards, executed seamless schema/code migrations, and eliminated operational downtime.
- **Data Privacy & DPDP Governance:** Spearheaded institutional compliance with India's Digital Personal Data Protection (DPDP Act 2023), implementing role-based access controls and PII anonymization across healthcare pipelines.
- **Technical Leadership & Enablement:** Directed end-to-end IT hardware provisioning (tablets, workstations, field devices) and trained clinical/program teams to achieve 100% digital adoption.

Data Analyst – Consultant / Project-Based **May 2024 – April 2026**
Bhagwati Udyog | Greater Noida, Uttar Pradesh, India
Engagement: Supply Chain Analytics, Predictive Demand Forecasting & ERP Inventory Modernization

- **Raw Material & Inventory Dashboard:** Engineered a centralized Power BI dashboard to monitor daily consumption and real-time stock levels, successfully reducing overall inventory carrying costs by 18%.
- **Predictive Demand Forecasting:** Built a predictive time-series model using Python (Prophet/ARMA) to forecast seasonal demand, optimizing raw material procurement and reducing component stockouts by 12%.
- **Workflow Automation:** Automated data ingestion pipelines using Python (Pandas) and SQL to clean inventory logs, eliminating 12 hours of manual entry per week and improving accuracy to 99%.

Marketing Data Analyst – Internship **Jul 2023 – Dec 2023**
Wunderman Thompson Studios | Gurgaon, India
Project: Campaign Analytics & ROI Optimization

- **Marketing ROI & Attribution (+18% ROI):** Integrated cross-channel CRM and marketing automation datasets from Salesforce and Marketo using SQL and Python, identifying high-LTV customer segments that boosted campaign ROI by 18%.
- **Statistical Experimentation (+22% Conversion):** Designed and evaluated multivariate A/B testing frameworks and performed deep Exploratory Data Analysis (EDA), validating creative and funnel optimizations that lifted lead conversion by 22%.
- **Automated BI Reporting (-30% Turnaround):** Built interactive Power BI executive dashboards with automated data refreshes, slashing client reporting turnaround times by 30%

EDUCATION

Diploma in Applied Artificial Intelligence Lab | WorldQuant University, U.S **Sept 2025 – Dec 2025**
Diploma in Applied Data Science Lab | WorldQuant University, U.S **Mar 2025 – Jun 2025**
Master of Computer Applications (MCA) | Vivekananda Institute of Professional Studies (VIPS), Delhi | CGPA: 9.5/10 **May 2021 - July 2023**
Bachelor of Computer Applications (BCA), Trinity Institute of Professional Studies (TIPS), Delhi | CGPA: 8.7/10 **Aug 2018 - July 2021**

INDEPENDENT PROJECT

Personal Finance Dashboard | Power BI, SQL, Data Modeling **Aug 2024 – Oct 2024**

- Engineered an end-to-end ETL pipeline using Python (Pandas) and SQL Server to ingest, cleanse, and normalize messy bank statements into a structured Data Warehouse.
- Automated transaction categorization, significantly reducing manual data processing by 90%.
- Delivered a unified view of spending patterns through an interactive dashboard that empowered users to optimize budget allocations and track financial health.

Environmental Forecasting (Delhi AQI) | Python, Time-Series Analysis, EDA **Dec 2024 – Jan 2025**

- Tackled a complex environmental issue by analyzing years of historical Air Quality Index (AQI) data for Delhi to uncover deep-rooted seasonal pollution patterns.
- Built and validated a predictive time-series model (Prophet/ARIMA) to accurately forecast future pollution spikes based on historical trends and cyclical environmental factors.
- Visualized high-dimensional environmental predictions into accessible graphs, demonstrating the ability to use data storytelling for strategic planning..

Real Estate Price Prediction Model (Gurugram) | Python, Regression Analysis **Jan 2025 – Feb 2025**

- Built a predictive pricing algorithm for the highly volatile Gurugram real estate market by analyzing historical property data, location features, and market trends.
- Developed an accurate regression model capable of estimating fair property valuations, providing actionable insights for potential real estate investments.

AI-Driven Healthcare Diagnostic Tool | Python, Random Forest, Logistic Regression **Mar 2025 - Apr 2025**

- Designed a predictive machine learning model using detailed patient health metrics to identify early risk factors for strokes.
- Evaluated multiple algorithms (Random Forest, Logistic Regression) to optimize diagnostic accuracy and minimize false negatives in a critical healthcare context.
- Highlighted the most critical health indicators driving stroke risk, translating complex medical data into interpretable technical insights.

Financial Risk & Fraud Detection Engine | Python, Scikit-Learn, SMOT **Sep 2025 - Nov 2025**

- Built a machine learning classification model to detect fraudulent credit card transactions within highly imbalanced financial datasets.
- Applied SMOTE (Synthetic Minority Over-sampling Technique) to balance the data, significantly improving the model's ability to catch rare anomalies.

RESEARCH PUBLICATIONS & HONOURS

- **Lead Author — Springer Publication (Jul 2023):** "Empowering the Visually Impaired: A Sustainable ML-Based Currency Recognition System" — Designed a lightweight TensorFlow Lite Computer Vision architecture; published in Sustainable Development through Machine Learning, AI and IoT (Springer).Data Analyst Certification
- **Co-Author — Springer Publication (Jul 2023):** "Advancing Sustainable Development Through Automation: An Advanced Health Monitoring System" — Engineered a multi-parameter ensemble ML system for predictive health analytics (Springer).Data Processing and Visualisation
- **Institutional Excellence Award:** Spearheaded the institutional data analytics framework and executive presentations for NAAC & NBA accreditation audits at VIPS, Delhi.Introduction to Database Systems

ACHIEVEMENTS AND CERTIFICATES

- **SQL (Intermediate) Certificate |** HackerRank (Feb 2026)
- **Data Analyst Certification |** OneRoadmap (Jun 2025)
- **AI and Data Scientist Certification |** OneRoadmap (Jul 2025)
- **Data Processing and Visualisation |** Ministry of Electronics and Information Technology (Aug 2024)
- **The Complete Python Bootcamp |** Udemy (Apr 2024)
- **Cloud Computing Fundamentals |** NPTEL (Apr 2023)
- **Conference Contributor |** International Conference on Sustainable Development (Jul 2023)